

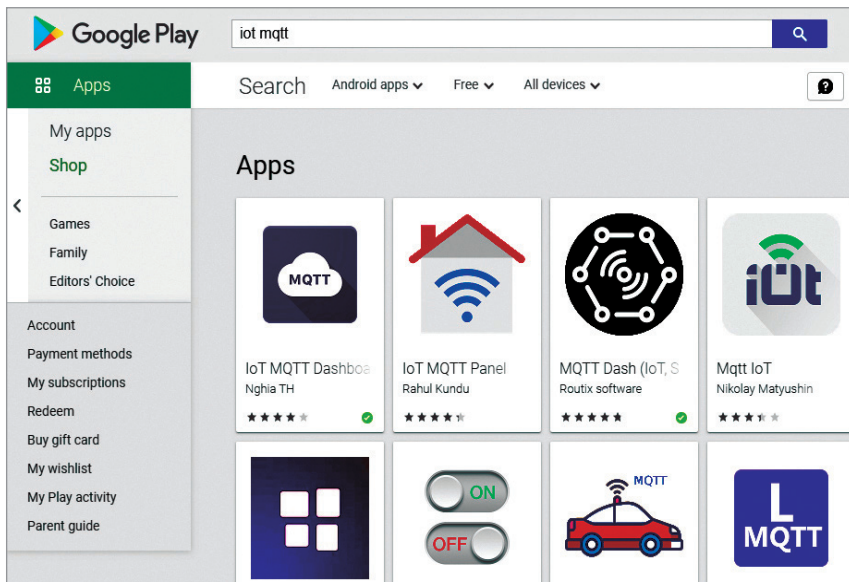


Fig. 4: IoT MQTT apps on Google Play Store

board on Google Play Store, as shown in Fig. 4. Such IoT MQTT applications communicate directly with the MQTT broker platforms so that the IoT apps, tablets, and cloud brokers can interact.

DIoTY

URL: <http://www.dioty.co>

DIoTY is another cloud based MQTT broker application with the free account that can be used to access IoT gadgets, as shown in Fig. 5. It provides the scripts so that MQTT broker connections can be rendered to various programming environments.

The following programming platforms can be used for communication between MQTT broker and smart gadgets for publishing as well as subscribing the data and signals:

- C#
- Java
- Python
- Arduino
- Node.JS
- PHP
- Go-Lang
- NodeMCU

Paho Client is one of the best libraries in Python used for communication and interfacing the devices. For implementation using Python, the package 'paho-mqtt' is installed as follows:

```
$ pip install paho-mqtt
```

The following is the code snippet that can be integrated with IoT based devices and gadgets:

```
import paho.mqtt.client as mqtt
# Define event callbacks
def on_connect(client, mySignalData, CN):
    if CN == 0:
        print("Connected successfully.")
    else:
        print("Connection failed. CN= "+str(CN))
    def on_publish(client, mySignalData, MyString):
        print("Message "+str(MyString)+" published.")
def on_subscribe(client, mySignalData, MyString, granted_qos):
    print("Subscribe with MyString "+str(MyString)+" received.")
def on_message(client, mySignalData, msg):
    print("Message received on topic "+msg.topic+" with QoS "+str(msg.qos)+" and payload "+msg.payload)
mqttclient = myMQTT.Client()
# Assign event callbacks
```

Prominent MQTT Brokers for IoT Based Communications	
MQTT applications	URL
Cloud MQTT	cloudmqtt.com
Mosquitto	mosquitto.org
EMQ X	emqx.io
FLESPI	flespi.com
Bevywise MQTTBroker	bevywise.com/mqtt-broker
M2MQTT	github.com/eclipse/paho.mqtt.m2mqtt
Mosca	github.com/mcollina/mosca
Paho MQTT	eclipse.org/paho
Rabbit MQ	rabbitmq.com
Thingstream	thingstream.io
VerneMQ	vernemq.com
HiveMQ	hivemq.com
MOQUETTE	github.com/moquette-io/moquette
wolfMQTT	wolfssl.com/products/wolfmqtt
EMQTTD	emqtt.io
NET-MQTT	hackage.haskell.org/package/net-mqtt
FLUUX	mqtt.fluux.io

using the current Google Account.

The key features of Cloud-MQTT broker are:

- 24 × 7 availability of MQTT broker
- Managed Mosquitto servers in the cloud
- Mosquitto implements the MQ Telemetry Transport protocol
- Lightweight methods for mes-

saging using a publish/subscribe message queuing model

The authentication details from CloudMQTT can be used in the IoT hardware or smartphones so that a real-time connection is established between CloudMQTT and gadgets, as shown in Fig. 3.

A variety of Android applications are available at the IoT MQTT dash-